GLUCOTWIN · TEAM AETHAN · AMITY UNIVERSITY MUMBAI

GlucoTwin

Multimodal Clinical Digital Twin for Type 2 Diabetes

2-Hour Ahead Glucose Spike Prediction & Closed-Loop Counterfactual Intervention Simulation



TEAM AETHAN — AMITY UNIVERSITY MUMBAI

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Reactive care sees the average. Patients live through the spikes.

Why retrospective quarterly HbA1c checks fail to prevent daily postprandial vascular damage

EPIDEMIC & PHENOTYPE

101+ million adults

India lives with Type 2 Diabetes, plus 136M pre-diabetic.

Asian-Indian phenotype: higher visceral adiposity and severe insulin resistance at lower BMI (23–25 kg/m²).

High triglycerides (>150 mg/dL), low HDL, and TCF7L2 (rs7903146) / PPARG (rs1801282) risk alleles compound risk.

WHY CURRENT CARE FAILS

The rear-view mirror

Quarterly HbA1c shows a 90-day average while hiding violent daily spikes (>180 mg/dL).

Repeated postprandial spikes trigger endothelial oxidative stress → retinopathy, nephropathy (eGFR decline), and neuropathy.

Static EHR, genetics, and wearable sensors remain siloed. Standard CGMs beep only after glucose crosses 180 mg/dL.

GLUCOTWIN · PROACTIVE CARE

Forecast before threshold

Fuses 53 Static EHR + Polygenic features with 54 Dynamic Wearable sensor channels.

Predicts >180 mg/dL excursions up to 120 minutes in advance (0.9925 ROC-AUC).

Doctors test walk duration, carb load, and medication dose before acting. Verified prescriptions reach the companion screen.

Six blocks turn multimodal signals into clinical action.

From Static EHR + 5-Minute Wearable IoT Streams to Real-Time Clinical Decision Support

01

Dual Data Streams

Static EHR + polygenic features alongside dynamic wearable / IoT streams.

02

Leak-Free Feature Pipeline

Patient-level partitioning and training-only scalers and encoders.

03

Multimodal AI Ensemble

Attention-LSTM + XGBoost with calibrated fusion.

04

Digital Twin Engine

Physiological state, risk triage, and counterfactual simulation.

05

FastAPI Microservice

Pre-fused tensors evaluated in <15ms.

06

Next.js Clinical Workstation

Physician, Patient Companion, and Cohort & AI Validation views.

Two streams. One patient-level split. Zero leakage.

STREAM 1 · STATIC / HISTORICAL EHR · 53 FEATURES

200 Synthetic Indian T2D Patients (P001–P200), ages 26–78, 8 Indian cities. HbA1c, Fasting BG, BP, LDL, HDL, Triglycerides, TG/HDL Ratio, eGFR, Creatinine. TCF7L2 (rs7903146), PPARG (rs1801282), Polygenic Risk Score (GRS); diabetes duration, ICMR BMI tier, medications, 5 comorbidities.

STREAM 2 · DYNAMIC WEARABLE / IOT · 54 FEATURES · 1.728M ROWS

30 days at 5-minute intervals = 8,640 readings per patient. CGM: interstitial glucose + velocity, acceleration, MAGE, CV%; HRV: RMSSD (ms), Heart Rate (bpm), autonomic stress index. Sleep stages (Awake / Light / Deep / REM), sleep quality, Steps/5m.

FUSION · 24 × 5 min — 2-hour lookback: X_temporal (N, 24, 54) + X_static (N, 53). Target: binary spike (>180 mg/dL within next 24 steps / 2 hours) + trajectory.

Partition	Patients	Windows	Spike rate
Training	140 · P001–P140	201,180	71.1%
Validation	20 · P141–P160	28,740	67.7%
Held-out test	40 · P161–P200	57,480	67.9%

0% patient leakage · Scalers & encoders fit strictly on the 140 training patients.

Sequence intelligence meets clinical feature interactions.

Attention-LSTM + Tabular XGBoost Ensemble

BRANCH A · WEIGHT 0.60

PyTorch Attention-LSTM

Temporal input: (Batch, 24, 54). Two-layer LSTM with 128 hidden units, LayerNorm, dropout (0.3), and glucose velocity / HRV lag capture.

Temporal self-attention learns softmax weights α_1..α_24 across the 2-hour window for clinician explainability.

Static EHR MLP: 53 → 64 → 32. Class-weighted BCE + cosine annealing.

BRANCH B · WEIGHT 0.40

Tabular XGBoost

485 engineered features: 432 multi-scale temporal summary statistics plus 53 Static EHR & polygenic risk features.

Last, mean, std, min, max, delta, linear slope, and early / late window means.

Histogram GBDT, max_depth=6, lr=0.05, with cost-sensitive positive class scaling.

Captures sharp cross-modal rules (e.g., post-meal slope + TCF7L2 TT genotype + high HbA1c).

FUSION · ATTRIBUTION · RUNTIME

Soft-Voting Rule

P_ens = 0.60 · P_LSTM + 0.40 · P_XGB.

Inter-model concordance computes confidence (0.50–0.99) from branch agreement.

Decomposes risk into signed ±mg/dL drivers tagged Wearable (Dynamic) vs. EHR (Static).

Pre-fused tensors evaluated in <15ms via FastAPI.

The branches complement one another: kinetic representation learning plus non-linear clinical tree interactions.

05 — EMPIRICAL VALIDATION

Held-out validation is precise on 40 unseen patients.

57,480 unseen 2-hour windows · P161–P200 · zero patient overlap

Evaluation Metric	Attention-LSTM (0.60) · XGBoost (0.40) · GlucoTwin Ensemble	Clinical Interpretation
ROC-AUC	0.9895 · 0.9937 · **0.9925**	Strong rank-order discrimination
PR-AUC (Avg Precision)	0.9951 · 0.9971 · **0.9965**	Robust precision across thresholds
Overall Accuracy	93.75% · 95.63% · **95.07%**	Consistent across 40 unseen patients
Sensitivity (Recall)	92.80% · 95.63% · **94.55%**	Catches 94.6% of impending spikes
Specificity	95.77% · 95.63% · **96.16% (Best)**	Minimizes clinician false-alarm fatigue
Precision (PPV)	97.89% · 97.89% · **98.12% (Best)**	98.1% of triggered alerts are true spikes
Brier Calibration Score	0.0443 · 0.0310 · **0.0357**	Well-calibrated probability estimates

HELD-OUT TEST CONFUSION MATRIX · 57,480 WINDOWS

True Positives (Spikes Caught): **36,922** · True Negatives (Safe Windows): **17,722**

False Positives (False Alarms): **708 (1.2%)** · False Negatives: **2,128** (borderline ~181 mg/dL)

CLINICAL VALUE

Combining Attention-LSTM and XGBoost achieves the highest Specificity (96.16%) and Precision (98.12%) on unseen patients, directly preventing physician alert fatigue.

Forecast the state, triage the risk, then test the intervention.

Hybrid ML + Mechanistic Physiological Compartment Modeling for “What-If” Prescriptions

01

01 · Virtual Patient Trajectory

Personalized kinetic calibration derives insulin resistance and hepatic clearance multipliers from HbA1c, BMI, duration, and GRS.

24-step forward projection forecasts the next 120 minutes and pinpoints the exact minute glucose crosses 180 mg/dL.

02

02 · ADA / RSSDI Risk Triage

Multi-factor score (0–100) integrates immediate CGM level, 2h ensemble spike probability, glucose velocity (mg/dL/min), and HbA1c.

Critical 75–100: immediate physician review. High 50–74: proactive counterfactual nudge. Moderate 25–49: automated lifestyle guidance. Low 0–24: routine surveillance.

03

03 · Parametric “What-If” Simulator

Hybrid mechanistic counterfactuals avoid observational confounding by indication.

Doctor controls: Post-Meal Brisk Walk (0–45m), Meal Carb Load (0–75g), and Scheduled Medication Dose.

Walk models non-insulin GLUT4 muscle glucose disposal (-35 to -55 mg/dL); carb load models gastric absorption surge; dose models sensitizer / secretagogue dampening.

The physician moves from triage to simulation to verified action.

Clean Hospital EHR Light Theme · Next.js · TypeScript · Tailwind CSS · Recharts

COHORT PRIORITY TRIAGE

200 patients, ranked

Automatically ranks patients by 2h spike risk with instant search and tier filters: All / Critical / High / Moderate.

STATIC EHR + GENETIC CARD

HbA1c, Fasting BG, BP, eGFR/Cr, LDL/TG, active regimen, comorbidities, and TCF7L2 / PPARG / GRS biomarkers.

SYNCHRONIZED PHYSIOLOGICAL CHART

Observed → forecast → simulate

Top panel: 6h observed CGM trace, 70–180 mg/dL green target zone, 2h red forecast, and green counterfactual overlay.

Bottom panel: synchronized wearable HRV RMSSD (ms), Heart Rate (bpm), and 5-minute Step Count bars on the exact same time axis.

INTERACTIVE COUNTERFACTUALS

Simulation in <15ms

Sliders: Post-Meal Walk (0–45 min), Meal Carbohydrate Load (0–75 g), and Scheduled Medication Dose.

Real-Time Peak Delta (Δ mg/dL) recomputes via POST /api/patients/{id}/simulate.

“Dispatch Nudge to Patient” sends the verified intervention to the Patient Companion View.

Signed ±mg/dL attribution bars and a 10-step LSTM Temporal Attention bar chart explain the result.

The loop closes when the patient acts and the system verifies.

From doctor simulation to patient action and cohort governance

VIEW 2 · PATIENT CLOSED-LOOP COMPANION APP

Verified intervention → completed action

Live 5-min sync status for CGM Patch, Smartwatch PPG (HRV RMSSD & HR), and Pedometer & Sleep Stage tracker.

Displays the physician-verified intervention, comparing Unmitigated 2h Peak vs. Target Peak With Action (-Δ mg/dL).

Patient taps “Mark Walk / Dose Completed,” updating the Physician Workstation: “✓ Patient Completed Action.”

Patient-friendly GLUT4 education explains how a 20-min post-meal walk clears glucose without extra insulin.

VIEW 3 · COHORT SURVEILLANCE & AI VALIDATION

Governance at population scale

Population Surveillance Strip shows live risk distribution across all 200 patients and 1.728M wearable rows: GET /api/stats/overview.

Held-Out Test Set Leaderboard compares Attention-LSTM (0.60), XGBoost (0.40), and GlucoTwin Ensemble: GET /api/stats/evaluation.

Confusion Matrix & Safety Verification: TP (36,922), TN (17,722), FP (708), FN (2,128).

Complete audit of the 53 Static EHR + 54 Dynamic Wearable features.

A bidirectional adherence loop connects action, outcome, and clinical oversight.

Proactive diabetes care can scale from sandbox to Indian hospitals.

DIRECT PATIENT BENEFIT

2-hour lead time: alerts arrive while glucose is still 135–155 mg/dL, before vascular toxicity occurs.

Behavioral micro-nudges: a 20-min post-meal walk drops peak glucose by 35–55 mg/dL via GLUT4 uptake.

Organ protection: higher Time-in-Range (70–180 mg/dL) protects kidneys (eGFR), retina, nerves, and coronary arteries.

HEALTHCARE BUSINESS IMPACT

Solves India's endocrinologist shortage (~1 per 35,000 patients).

>75% reduction in triage time: one clinician can oversee 200+ active CGM patients.

Actionable 2h nudges prevent CGM drop-off after 14 days and support subscription CGM retention.

37 automated unit tests and <15ms API latency.

PRODUCTION ROADMAP

Phase 1 · Complete

200-patient synthetic sandbox, Attention-LSTM + XGBoost ensemble, 3-view clinical workstation.

Phase 2 · Hospital Pilot

Ayushman Bharat Digital Mission (ABDM) HL7 FHIR R4 integration for live EHR ingestion.

Phase 3 · Edge & Wearables

Direct webhooks for Abbott FreeStyle Libre, Apple HealthKit & Google Health Connect + 7-day Bayesian twin personalization.